

JANE DOE - CURRICULUM VITAE

RESEARCHER IN STATISTICS

Statistical methods, Machine learning

Sparse models · Multi-omics integration · Reproducible research



Born January 1, 1990
Example citizen

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BRIEF SUMMARY OF ACTIVITIES

<i>Production</i>	2 papers, 2 talks.
<i>Students</i>	1 PhD, 1 Master.
<i>Teaching</i>	Statistics in Action with R, Convex Optimization.

EXPERIENCE

2011–	RESEARCHER, EXAMPLE INSTITUTE
2008–2011 <i>supervisor</i>	POSTDOC, EXAMPLE UNIVERSITY Dr. Example

EDUCATION

2015 <i>Title</i>	HABILITATION IN MATHEMATICS <i>Sparse methods for complex data</i>
2003	PHD IN APPLIED MATHEMATICS

SCIENTIFIC ACTIVITIES

GRANTS

ON GOING PROJECTS

2023–2027 <i>Partners</i>	DISCERN – DISCOVERING THE CAUSES OF A POORLY UNDERSTOOD CANCER 20 partners
<i>Description</i>	A five-year consortium project combining statistical genomics, imaging, and epidemiological cohort data to identify early biomarkers of a poorly understood cancer subtype. My group leads the statistical modelling work package, developing sparse graphical models for multi-omics integration across the twenty partner sites.
<i>Support</i>	Horizon Europe

PAST

2019–2023 <i>Support</i>	ECONET – ADVANCED STATISTICAL MODELLING OF ECOLOGICAL NETWORKS French National Research Agency (ANR)
2017–2019 <i>Leader</i>	SEARS – SAMPLING STRATEGIES AND NETWORK ANALYSIS Mathieu Thomas

OTHER GRANTS

2010–2012	OLDER GRANT, LISTED ONLY IN THE LONG VERSION
<i>Support</i>	French National Research Agency (ANR)

ACTIVITIES

SCIENTIFIC COMMITTEES

	COUNCILS
<i>since 2022</i>	Member of the Foundation Council
2020–2024	PHD COMMITTEE (full list)
<i>role</i>	Reviewer for 12 PhD theses

STUDENTS

PHD STUDENTS

2021–2024	JANE ROE – SPARSE STATISTICAL METHODS FOR LARGE-SCALE MULTI-OMICS DATA
<i>Co-supervisor</i>	Prof. Example
<i>Description</i>	Developed sparse graphical model estimators for integrating heterogeneous omics data sources, with applications to the DISCERN consortium. Defended in 2024; now a postdoctoral researcher at Example University.

MASTER STUDENTS

2023	JOHN ROE – POISSON-LOGNORMAL MODELS FOR COUNT DATA
<i>Co-supervisor</i>	Dr. Example

TEACHING

Approximately 2000 hours of teaching. Lecturer at Example University from 2015 to 2024.

2020–24	STATISTICS IN ACTION WITH R
<i>Msc</i>	Probabilistic models, data analysis, R programming
2017	CONVEX OPTIMIZATION (12h course)
<i>Msc</i>	Gradient methods, Newton method, Proximal methods

SCIENTIFIC PRODUCTIONS

PUBLICATIONS

SELECTED PUBLICATIONS

- [JP1] J. Doe and J. Roe. “A Toy Example for the quarto-cv-blocks Extension”. In: *Journal of Reproducible Examples* 1 (2023), pp. 1–10. DOI: [10.0000/example.2023.001](#).
- [JP2] J. Doe. “Demonstrating a Fictitious Conference Paper”. In: *Proceedings of the Example Conference*. 2021, pp. 42–50.

SELECTED TALKS

- [CT1] J. Doe. “Reproducible CVs with Quarto”. In: *Example Workshop on Reproducible Documents*. Example City, 2022.
- [CT2] J. Doe and J. Roe. “A Fictitious Talk for the quarto-cv-blocks Example”. In: *Proceedings of the Other Example Conference*. 2020, pp. 12–20.

RESEARCH PROJECTS

DISCERN

As part of the DISCERN project, my group develops sparse graphical model estimators for integrating multi-omics data. Given a sample covariance matrix S , we estimate a sparse precision matrix $\hat{\Theta}_\lambda$ by solving

$$\hat{\Theta}_\lambda = \arg \min_{\Theta \succ 0} -\log \det \Theta + \text{tr}(S\Theta) + \lambda \|\Theta\|_1.$$

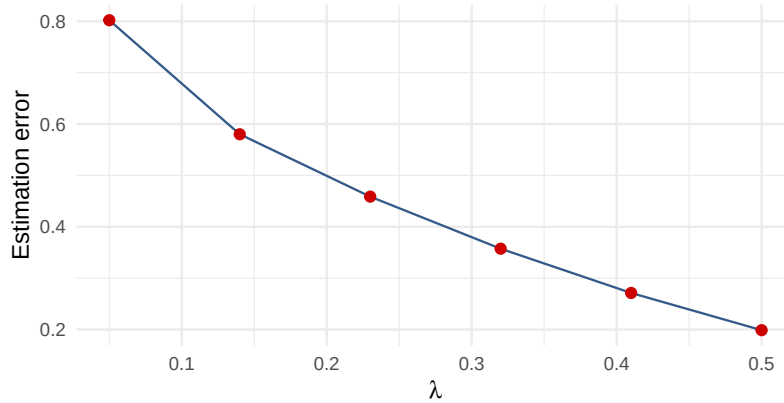


Figure 1: Estimation error as a function of the sparsity penalty

Increasing the penalty λ trades estimation error for a sparser, more interpretable graph, see Figure 1.

ANOTHER PROJECT

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See Table 1.

Lambda	Error	Sparsity
0.05	0.79	12%
0.20	0.42	48%
0.50	0.19	81%

Table 1: Estimation error and graph sparsity for a few penalty values